

Smart Traffic Lights Using Local AI

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Abstract—Smart traffic-light systems are being studied as a way to reduce vehicle waiting time and respond more quickly to changing road conditions. Traditional traffic lights generally operate using fixed timers and are simple and inexpensive, but they cannot easily respond when traffic suddenly becomes crowded on one road. Local artificial intelligence (AI) systems can process traffic information closer to the road and make faster decisions without depending entirely on a central cloud server. This paper presents a conceptual architecture for smart traffic lights using local AI, where cameras capture traffic conditions, local computers process the information, and traffic-light signals are adjusted accordingly. The proposed system also identifies emergency vehicle detection as an important research gap, particularly for automatically creating a clear route for ambulances.

Keywords— Smart Traffic Lights, Local AI, Artificial Intelligence, Traffic Management, Machine Vision, Emergency Vehicles.

I. INTRODUCTION

Traditional traffic-light systems generally operate using fixed timers. Although such systems are simple and inexpensive, they have limited ability to react to sudden changes in traffic density. Smart traffic-light systems attempt to overcome this limitation by using information collected from the road.

Recent research has investigated artificial intelligence and machine-vision techniques for adaptive traffic signal control [1], [2]. Instead of changing signals only according to a fixed clock, an intelligent system can use information about the number of vehicles and traffic conditions to support signal decisions.

Cloud-based AI systems can process camera information on a central server. However, such systems depend on communication networks and may introduce communication delays. Local smart-camera systems move some processing closer to the road. This approach can reduce the amount of information transmitted over the internet and may support faster decisions.

II. LITERATURE REVIEW

AI-based traffic signal control has received increasing research attention. Rasheed et al. reviewed deep reinforcement learning approaches for traffic signal control and discussed their potential for adaptive traffic management [1].

Machine-vision-based approaches have also been investigated for adaptive traffic-light control. Gorodokin et al. studied the optimization of adaptive traffic-light control modes using machine vision [2].

These studies demonstrate the potential of using traffic information and intelligent processing for adaptive signal control. However, the specific implementation of local AI for rapid emergency-vehicle priority remains an important area for further investigation.

III. CLOUD AI AND LOCAL AI

Cloud-based AI systems send camera information to a central server for processing. This can provide access to greater computing resources, but it requires network connectivity and may introduce communication delays.

Local AI systems perform some of the processing near the traffic signal. A small local computer can receive camera information, estimate traffic conditions, and send a short control instruction to the traffic-light controller.

The main conceptual difference can be summarized as follows:

- **Cloud AI:** Camera data is transmitted to a central server for processing.
- **Local AI:** Traffic information is processed close to the road.
- **Hybrid approach:** Local processing is used for rapid decisions while selected information is sent to a cloud server for storage and wider analysis.

IV. SYSTEM ARCHITECTURE

The proposed architecture consists of four major stages:

- 1) **Street Cameras:** Capture information about vehicles and traffic conditions.
- 2) **Smart Pole Computers:** Process the camera input locally and make rapid traffic decisions.
- 3) **City Cloud Server:** Store selected information and support wider traffic analysis.
- 4) **Traffic Lights:** Receive control decisions and change their signals accordingly.

The overall flow is:

Street Cameras → Smart Pole Computers → City Cloud Server → Traffic Lights

V. SYSTEM COMPARISON

The three conceptual systems can be compared using decision time, internet-data usage, and setup cost.

TABLE I
COMPARISON OF TRAFFIC-LIGHT SYSTEMS

System	Decision Time	Internet Data	Setup Cost
Basic Timed Lights	30 s	Zero	\$200
Cloud AI Systems	2 s	High	\$2,500
Local Smart Cameras	0.5 s	Low	\$1,200

The values in Table I are the conceptual comparison values provided in the workshop material and should not be interpreted as results from an experimental implementation.

VI. RESEARCH GAP

Existing traffic systems can work effectively for regular daily traffic, particularly when traffic patterns are predictable. However, an important emergency-response problem remains.

Many traffic systems do not have a built-in mechanism for automatically creating a clear route for a speeding ambulance. Consequently, there is a need for a smart traffic system that can detect an emergency vehicle and coordinate nearby traffic signals quickly.

A possible direction for future research is to combine local AI-based vehicle detection with emergency-vehicle recognition. When an authorized emergency vehicle is detected, nearby signals could be coordinated to provide a suitable route while maintaining safety for other road users.

VII. PROPOSED EMERGENCY VEHICLE CONCEPT

The proposed extension focuses on emergency vehicles, particularly ambulances. The conceptual process is:

- 1) Cameras monitor the traffic lanes.
- 2) Local AI processes the camera information.
- 3) The system detects an emergency vehicle.
- 4) Nearby traffic signals are identified.
- 5) The system determines an appropriate signal sequence.
- 6) Traffic signals are coordinated to support the emergency vehicle's movement.

This concept requires further testing to determine its accuracy, response time, safety, and reliability under real-world traffic conditions.

VIII. ADVANTAGES

The proposed local-AI approach has several potential advantages:

- Faster local processing of traffic information.
- Reduced dependence on continuous internet connectivity.
- Lower amount of traffic information transmitted to the cloud.
- Ability to respond to changing traffic conditions.
- Potential support for emergency-vehicle priority.

IX. LIMITATIONS

The conceptual system also has several limitations:

- Camera-based systems may be affected by poor visibility, weather, or camera obstruction.
- Local AI hardware requires appropriate processing capability.
- Emergency-vehicle detection must be highly reliable before being used in real traffic.
- Incorrect signal decisions could create safety risks.
- The proposed architecture requires practical testing before real-world deployment.

X. CONCLUSION

Smart traffic-light systems provide a potential approach for improving traffic management by using information from the road rather than relying only on fixed timers. Local AI can move part of the processing closer to the traffic signals, which may reduce communication delay and internet-data requirements.

The proposed architecture uses street cameras, local smart-pole computers, a cloud server, and traffic lights. An important research gap is the automatic detection of emergency vehicles and coordination of nearby traffic signals to create a clear route. Future research should focus on developing and experimentally evaluating such a system under different traffic and environmental conditions.

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